

BAYESIAN CONSISTENCY FOR A CLASS OF REGRESSION PROBLEMS

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Summary: We consider a general result for establishing posterior consistency for a class of Bayesian regression models. To date, no such general results exist. Consistency will be for a random function which corresponds to a mean regression function for a one parameter family of density functions.

1. Introduction

Bayesian consistency is currently an active area of research. A recent survey of the literature is provided by Ghosh and Ramamoorthi (2003). The strong message coming from recent papers is that for nonparametric priors a support condition and “something more” is required to establish Hellinger consistency for both the sequence of posterior distributions and corresponding sequence of predictive density functions. The “something more” is to prevent densities, within the support of the prior, from tracking the data too closely. Such densities assign large likelihood values and hence take up large amounts of mass in the posterior.

The support condition is that the prior probability measure puts positive mass on all Kullback–Leibler neighbourhoods of the true density function, say

f_0 . That is, if Π denotes the prior, then

$$\Pi \{f : d_K(f, f_0) < \varepsilon\} > 0$$

for all $\varepsilon > 0$. Here $d_K(f, f_0)$ denotes the Kullback–Leibler divergence between f and f_0 . We shall refer to this as the Kullback–Leibler property for Π . A variety of the extra conditions sufficient for Hellinger consistency can be found in Barron et al. (1999), Ghosal et al. (1999) and Walker (2003a).

In this paper we provide a new sufficient condition for posterior Hellinger consistency. We will then use this result to work on consistency for a class of regression problems. It is convenient at this point to introduce the notation. We define

$$f_n = \int f \Pi_n(df)$$

to be the predictive density function and

$$\Pi_n(df) = \frac{\prod_{i=1}^n f(Y_i) \Pi(df)}{\int \prod_{i=1}^n f(Y_i) \Pi(df)}$$

to be the posterior distribution. Here $Y_1, \dots, Y_n$ are independent and identically distributed from f_0 . We write

$$H(f, f_0) = \left\{ \int \left(\sqrt{f} - \sqrt{f_0} \right)^2 \right\}^{1/2} = \left[2 \left\{ 1 - \int \sqrt{(f_0 f)} \right\} \right]^{1/2}$$

as the Hellinger distance between f and f_0 , and we will also use the notation $h(f, f_0) = \frac{1}{2} H^2(f, f_0)$, and note that $h(f, f_0) \leq 1$. In fact,

$$h(f, f_0) = 1 - \int \sqrt{f f_0}$$

and we assume all densities are with respect to the Lebesgue measure. We define

$$I_n = \int R_n(f) \Pi(df)$$

where

$$R_n(f) = \prod_{i=1}^n f(Y_i)/f_0(Y_i).$$

Additionally,

$$f_{nA} = \int f \Pi_{nA}(df)$$

is the predictive density function with posterior distribution restricted, and normalized, to the set A and $L_{nA} = \int_A R_n(f) \Pi(df)$ with $L_{0A} = \Pi(A)$.

The identity

$$L_{n+1A}/L_{nA} = f_{nA}(Y_{n+1})/f_0(Y_{n+1}) \quad (1)$$

holds with

$$L_{1A} = \Pi(A) f_{0A}(Y_1)/f_0(Y_1)$$

where $f_{0A} = \int f \Pi_A(df)$ and Π_A is the prior restricted, and normalized, to the set A , assuming that $\Pi(A) > 0$.

In Section 2 we provide details of the regression problem to be considered. Section 3 provides a posterior consistency theorem for independent and identically distributed observations which is then adapted to provide a posterior consistency theorem for a class of regression models. Section 4 discusses a normal and binary regression problem and the paper concludes with a discussion in Section 5.

2. The regression problem

Consider a sample $(X_1, Y_1), \dots, (X_n, Y_n)$ where the X_i are $\mathfrak{X}$ valued random observations and the Y_i the corresponding response observations. For concreteness it can be assumed that $\mathfrak{X} \subseteq \mathfrak{R}$ and the X_i are independent and identically distributed with density $m(x)$ with respect to the Lebesgue measure.

The distribution of the responses is of a known form with parameter depending on the X_i ; that is,

$$Y_i|[X_i = x_i] \sim f(\cdot|\theta(x_i)),$$

where $\theta \in \Theta$ and Θ is a separable space of real valued functions on $\mathcal{R}$, with respect to the metric d between elements of Θ . For example, the space of continuous functions on $[0, 1]$, $\Theta = \mathcal{C}[0, 1]$, equipped with the sup-metric is, according to Billingsley (1968), complete and separable. Hence, for any $\delta > 0$ there exists a countable set $\{\theta_1, \theta_2, \dots\}$ such that Θ is covered by $\mathcal{A}_\delta = \{A_1, A_2, \dots\}$, where $A_k = \{\theta : d(\theta_k, \theta) < \delta\}$.

The function θ will play the role of the mean, so $E(Y|X = x) = \theta(x)$, and $f(\cdot|\theta)$ will be a known one-parameter family of density functions.

The prior distribution on Θ will be denoted by $\Pi(d\theta)$ and we assume there exists a true parameter value $\theta_0 \in \Theta$.

The problem has been extensively studied from a classical perspective. Perhaps the closest piece of work is by Yatracos (1989), who estimates θ consistently with respect to the L_1 distance. Yatracos (1989) assumes $Y_i|[X_i = x_i] \sim f(\cdot|x_i, \theta(x_i))$ and that Θ is a sup-norm compact infinite dimensional space of real valued functions on $\mathcal{X}$, and $\mathcal{X}$ is a compact set in $\mathcal{R}^d$ for $d = 1, 2$. Yatracos (1989) also assumes that the L_1 distance (equivalently the Hellinger distance) of the conditional densities $\|f(\cdot|x, \theta_1(x)) - f(\cdot|x, \theta_2(x))\|$ is of the order of $|\theta_1(x) - \theta_2(x)|$ for all $\theta_1, \theta_2 \in \Theta$ and $x \in \mathcal{X}$. We will also be making a similar assumption; that

(A1) For any $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\sup_x H(f(\cdot|\theta_1(x)), f(\cdot|\theta_2(x))) < \varepsilon$$

whenever $d(\theta_1, \theta_2) < \delta$.

(A2) For any $\delta > 0$ there exists an $\varepsilon > 0$ such that

$$\inf_x H(f(\cdot|\theta_1(x)), f(\cdot|\theta_2(x))) > \varepsilon$$

whenever $d(\theta_1, \theta_2) > \delta$.

From a Bayesian point of view, Diaconis and Freedman (1993) consider consistency for a binary regression problem,

$$P(Y = 1|X = x) = \theta(x)$$

where x is a sequence of 0's and 1's. Under a specific prior, balanced data and $\theta \neq \frac{1}{2}$, Diaconis and Freedman (1993) demonstrate that the posterior accumulates in L_2 neighbourhoods of θ_0 . This work was extended to normal regression in Diaconis and Freedman (1998) where they assume $E(Y|X = x) = \theta(x)$, $\text{Var}(Y|X = x) = 1$ and as with the binary regression model, x is a vector of binary covariates.

Shen and Wasserman (2001) consider a special regression problem given by

$$Y_i = \theta(X_i) + e_i$$

where the e_i 's are independent $N(0, 1)$ random variables, the X_i 's are independent uniform random variables from the interval $(0, 1)$. For a class of function of θ , based on an orthonormal basis in $L_2[0, 1]$, and using a mixture of priors, they provide rates of convergence for the posterior distribution accumulating in L_2 neighbourhoods of θ_0 .

Amewou-Atisso et al. (2003), on the other hand, consider the linear regression model

$$Y_i = a + bX_i + e_i$$

where the e_i 's are independent random variables from the unknown symmetric density function f . They establish sufficient conditions under which the posterior accumulates in suitable neighbourhoods of (f_0, a_0, b_0) .

To date there appear to be no general results for Bayesian consistency involving regression problems. This paper provides some general results.

3. Posterior consistency

To start we will consider consistency for independent and identically distributed observations. The result generalizes a result from Walker (2003b). Let us split the space of density functions into a countable disjoint union of sets A_k , where

$$A_k \subseteq A_k^* = \{f : h(f, f_k) < \varepsilon/2\}$$

and $\{f_1, f_2, \dots\}$ are a set of densities. This is possible because of the separability of the space of density functions with respect to the Hellinger distance.

PROPOSITION 1. Suppose Π puts positive mass on all Kullback–Leibler neighbourhoods of f_0 and there exists a $\frac{1}{2} \leq \alpha < 1$ such that

$$\sum_k \Pi(A_k)^\alpha < \infty;$$

then

$$\Pi_n(\{f : H(f, f_0) > \varepsilon\}) \rightarrow 0$$

for all $\varepsilon > 0$ with probability one.

PROOF. Appendix A1.

REMARK 1. When the prior is discrete, putting mass Π_k on the density f_k , the condition becomes the existence of $\alpha \in [\frac{1}{2}, 1)$ such that

$$\sum_k \Pi_k^\alpha < \infty.$$

This is essentially Theorem 15 from Barron (1988). It is somewhat surprising that the extension from discrete prior was not available with the techniques employed by Barron, which were the use of uniformly consistent tests. The result also generalises the finding of Walker (2003b) who along with Π putting positive mass on all Kullback–Leibler neighbourhoods of f_0 found that

$$\sum_k \Pi(A_k)^{1/2} < \infty$$

was sufficient for posterior Hellinger consistency. Indeed, in the next section we will restrict attention to $\alpha = \frac{1}{2}$.

3.1 Consistency for the regression problem

To ease the notation, we will define

$$\mathcal{L}_n(\theta) \equiv \prod_{i=1}^n f(Y_i | \theta(X_i)).$$

Then, for any set measurable set A , we have

$$\Pi\{\theta \in A | (X_1, Y_1) \dots (X_n, Y_n)\} \equiv \Pi_n(A) = \frac{\int_A \mathcal{L}_n(\theta) \Pi(d\theta)}{\int \mathcal{L}_n(\theta) \Pi(d\theta)}.$$

The goal of the paper is to establish conditions under which

$$\Pi_n(\{\theta : d(\theta_0, \theta) > \varepsilon\}) \rightarrow 0$$

with probability one, for all $\varepsilon > 0$. We will, as is usual with the Bayesian consistency literature (Barron et al., 1999; Ghosal et al., 1999; Walker, 2003a), consider separately the numerator and denominator; defining

$$I_n \equiv \int \{\mathcal{L}_n(\theta) / \mathcal{L}_n(\theta_0)\} \Pi(d\theta)$$

and

$$J_n(A) \equiv \int_A \{\mathcal{L}_n(\theta) / \mathcal{L}_n(\theta_0)\} \Pi(d\theta).$$

3.1.1 Denominator

The I_n is typically bounded below using a support condition. This is quite standard and the required condition dates back to Schwartz (1965). That is, if

$$\Pi \left(\left\{ \theta : \int \int \log \{ f(y|\theta_0(x)) / f(y|\theta(x)) \} f(y|\theta_0(x)) m(x) dy dx < \varepsilon \right\} \right) > 0$$

for all $\varepsilon > 0$ then $I_n > \exp(-nc)$ a.s. for all large n , for any $c > 0$.

We are then interested in understanding when $J_n(A) < \exp(-nb)$ a.s. for all large n , for some $b > 0$, when $A = \{ \theta : d(\theta_0, \theta) > \varepsilon \}$.

3.1.2 Numerator

To deal with the numerator, we have the identity

$$J_{n+1}(A) / J_n(A) \equiv f_{nA}(Y_{n+1}) / f_{n\theta_0}(Y_{n+1}),$$

where f_{nA} is the predictive density with posterior restricted to the set A ; that is,

$$f_{nA}(y) = \int f(y|\theta(X_{n+1})) \Pi_{nA}(d\theta)$$

and

$$\Pi_{nA}(d\theta) = \frac{\mathbf{1}(\theta \in A) \mathcal{L}_n(\theta) \Pi(d\theta)}{\int_A \mathcal{L}_n(\theta) \Pi(d\theta)}.$$

Also,

$$f_{n\theta_0}(y) = f(y|\theta_0(X_{n+1})).$$

Note that we also have

$$J_1(A) \equiv \Pi(A) f_{0A}(Y_1) / f_{0\theta_0}(Y_1),$$

where $f_{0A}(y) = \int f(y|\theta(X_1)) \Pi_A(d\theta)$ and $f_{0\theta_0}(y) = f(y|\theta_0(X_1))$. It is therefore quite easy to see that

$$E \left\{ J_{n+1}^{1/2}(A) | Y_1 \dots Y_n, X_1 \dots X_{n+1} \right\} = J_n^{1/2}(A) \{ 1 - h(f_{nA}, f_{n\theta_0}) \}. \quad (2)$$

THEOREM 1. *Let Θ be separable space of real valued functions with respect to the metric d . Let $X_1, X_2, \dots$ be independent covariates with*

identical density $m(x)$, and $Y_i|X_i = x_i \sim f(\cdot|\theta_0(x_i))$ where f is a known family of densities. If conditions (A1) and (A2) are satisfied, Π has the support condition of Section 3.1.1 and

$$\sum_{A_k \in \mathcal{A}_\delta} \sqrt{\Pi(A_k)} < \infty$$

for all $\delta > 0$ then Π_n is consistent.

PROOF. Appendix A2.

4. Illustrations

The basic idea in this section will be to look at specific models and establish whether (A1) and (A2) can be satisfied for a particular distance function d . For obvious reasons this will invariably be the sup-metric; $d(\theta_1, \theta_2) = \|\theta_1 - \theta_2\|_\infty = \sup_x |\theta_1(x) - \theta_2(x)|$. We will also look at the support condition. These will be done in Section 4.1 and 4.2 for the normal and Bernoulli models, respectively. Because the

$$\sum_{A_k \in \mathcal{A}_\delta} \sqrt{\Pi(A_k)} < \infty$$

condition is quite different we will look at this for the normal and Bernoulli models in Sections 4.3 and 4.4, respectively.

4.1 Normal model

Here we consider a normal regression model

$$f(y|\theta(x)) = N(y|\theta(x), \sigma^2)$$

for which σ is assumed known. Now, letting $f_\theta(\cdot)$ denote a generic $f(\cdot|\theta(x))$, we have

$$\begin{aligned}
h(f_{\theta_1}, f_{\theta_2}) &= 1 - \frac{1}{\sigma\sqrt{2\pi}} \int \exp \left\{ -\frac{1}{4\sigma^2} [(y - \theta_1(x))^2 + (y - \theta_2(x))^2] \right\} dy \\
&= 1 - \frac{1}{\sigma\sqrt{2\pi}} \int \exp \left\{ -\frac{1}{4\sigma^2} [2y^2 - 2y(\theta_1 + \theta_2) + \theta_1^2 + \theta_2^2] \right\} dy \\
&= 1 - \frac{1}{\sigma\sqrt{2\pi}} \int \exp \left\{ -\frac{1}{2\sigma^2} [y^2 - 2y\bar{\theta} + \bar{\theta}^2] \right\} dy \\
&= 1 - \frac{1}{\sigma\sqrt{2\pi}} \int \exp \left\{ -\frac{1}{2\sigma^2} [(y - \bar{\theta})^2 + \bar{\theta} - \bar{\theta}^2] \right\} dy \\
&= 1 - \exp \left\{ -\frac{1}{2\sigma^2} (\bar{\theta} - \bar{\theta}^2) \right\}.
\end{aligned}$$

Here $\bar{\theta} = \frac{1}{2}(\theta_1 + \theta_2)$ and $\bar{\theta}^2 = \frac{1}{2}(\theta_1^2 + \theta_2^2)$. Therefore,

$$h(f(\cdot|\theta_1(x)), f(\cdot|\theta_2(x))) = 1 - \exp \left\{ -(\theta_1(x) - \theta_2(x))^2 / (8\sigma^2) \right\}$$

and so if $d(\theta_1, \theta_2) = \|\theta_1 - \theta_2\|_\infty$ we have met the (A1) and (A2) conditions.

To consider the support property, we have

$$\begin{aligned}
&\int \int \log \{ f(y|\theta_0(x)) / f(y|\theta(x)) \} f(y|\theta_0(x)) m(x) dy dx \\
&= c \int \{ \theta(x) - \theta_0(x) \}^2 m(x) dx
\end{aligned}$$

for some constant $c > 0$, and so we require θ_0 to be in the L_2 support of

II. If $\mathcal{X}$ is $[0, 1]$ and $m(x)$ is flat then we can consider

$$\theta(x) = \sum_{j=1}^{\infty} \theta_j \psi_j(x)$$

where each θ_j has support on a subset of $(-\infty, +\infty)$ and the ψ_j form an orthonormal basis for $L_2[0, 1]$.

4.2 Bernoulli model

A binary regression problem also meets conditions (A1) and (A2) with the sup-metric. We have

$$f(y|\theta(x)) = \theta(x)^y \{1 - \theta(x)\}^{1-y}$$

with $y \in \{0, 1\}$. Then

$$\begin{aligned} h(f_{\theta_1}, f_{\theta_2}) &= 1 - \sum_{y=0}^1 \sqrt{f(y|\theta_1) f(y|\theta_2)} \\ &= 1 - \sqrt{(1-\theta_1)(1-\theta_2)} - \sqrt{\theta_1\theta_2} \end{aligned}$$

and it is easy to show (see Appendix A3) that

$$\frac{1}{2}|\theta_1 - \theta_2| \leq 1 - \sqrt{(1-\theta_1)(1-\theta_2)} - \sqrt{\theta_1\theta_2} \leq |\theta_1 - \theta_2|$$

for any $(\theta_1, \theta_2) \in [0, 1]^2$.

To meet the support requirement in this case one would like $0 < \theta_0 < 1$ and that θ_0 lies on the d_∞ support of Π ; that is

$$\Pi \left(\left\{ \theta : \sup_x |\theta(x) - \theta_0(x)| < \varepsilon \right\} \right) > 0$$

for all $\varepsilon > 0$. If $\Theta = \mathcal{C}[0, 1]$, the space of continuous, real valued functions on $[0, 1]$ with the sup-metric, then Θ is separable (Billingsley, 1968). One such countable, dense set consists of functions linear on each subinterval $[(i-1)/k, i/k]$, $i = 1 \dots k$ for some integer k and assume rational values at points i/k , $i = 0 \dots k$. See Billingsley (1968) for further details. Hence a prior can be put on such functions and full support will follow.

4.3 Normal model revisited

Here we consider again the normal regression model and now we take

$$\theta(x) = \sum_{l=1}^{\infty} \theta_l \psi_l(x),$$

where the θ_l are independent $N(0, \sigma_l^2)$ random variables (characterizing the prior) and the ψ_l are a set of basis functions on L_2 . We assume the ψ_j are such that if

$$\sum_{l=1}^{\infty} |\theta_{1l} - \theta_{2l}| < \delta$$

then $\|\theta_1 - \theta_2\|_\infty < \delta$. For example, by ensuring that $\sup_x |\psi_l(x)| \leq 1$ for all l . Hence, we can investigate the finiteness of

$$\lim_{M \rightarrow \infty} \sum_{n_1=-\infty}^{+\infty} \cdots \sum_{n_M=-\infty}^{+\infty} \prod_{l=1}^M P^{1/2}\{\theta_l \in (n_l \delta_l, (n_l + 1) \delta_l)\}$$

which holds if

$$\prod_{l=1}^{\infty} \sum_{n=0}^{\infty} P^{1/2}\{\theta_l \in (n \delta_l, (n + 1) \delta_l)\} < \infty.$$

Here δ_l is any sequence such that $\sum_l \delta_l \leq \delta$; for example, $\delta_l = \delta w_l / \sum_l w_l$ for any non-negative sequence w_l such that $\sum_l w_l < \infty$.

Now, letting z denote a standard normal random variable,

$$P\{\theta_l \in (n \delta_l, (n + 1) \delta_l)\} = P\{z \in (n \lambda_l, (n + 1) \lambda_l)\},$$

where $\lambda_l = \delta_l / \sigma_l$, and, for $n > 0$,

$$P\{z \in (n \lambda_l, (n + 1) \lambda_l)\} < c \lambda_l \exp\left(-\frac{1}{2} n^2 \lambda_l^2\right)$$

for some constant c . Therefore,

$$\begin{aligned} \sum_{n=0}^{\infty} P^{1/2}\{\theta_l \in (n \delta_l, (n + 1) \delta_l)\} &\leq 1 + c^{1/2} \frac{\lambda_l^{1/2}}{\exp(\lambda_l^2/4) - 1} \\ &\leq 1 + c^* \lambda_l^{1/2 - 2m} \end{aligned}$$

for any positive integer m . Here c^* represents another constant. Then

$$\sum_{l=1}^{\infty} (\sigma_l / \delta_l)^{2m-1/2} < \infty$$

is sufficient. We can take $\delta_l \propto l^{-1-r}$ for any $r > 0$ and so $\sigma_l \propto l^{-1-r}$ for any $r > 0$ is sufficient. This follows because we will have $\sigma_l / \delta_l = l^{-r}$ for some $r > 0$ and we can make m large enough to ensure that

$$\sum_{l=1}^{\infty} l^{-r(2m-1/2)} < \infty.$$

4.4 Bernoulli model revisited

Here we consider a model for θ given by

$$\theta(x) = \sum_{l=1}^{\infty} \theta_l g_l(x),$$

where the $g_l(x)$ are a set of basis functions on $[0, 1]$ and $\sum_l \theta_l = 1$. It is quite clear that if

$$\sum_{l=1}^{\infty} |\theta_{1l} - \theta_{2l}| < \delta$$

then $\|\theta_1 - \theta_2\|_{\infty} < \delta$. We are then interested in the finiteness of

$$\prod_{l=1}^{\infty} \sum_{n=1}^{1/\delta_l} P^{1/2} \{ \theta_l \in ((n-1)\delta_l, n\delta_l) \},$$

where $\sum_l \delta_l = \delta$. Now, for $n > 1$,

$$\begin{aligned} P\{\theta_l \in ((n-1)\delta_l, n\delta_l)\} &\leq P\{\theta_l \geq (n-1)\delta_l\} \\ &\leq \mu_l / \{\delta_l(n-1)\} \leq \mu_l / \delta_l, \end{aligned}$$

where $\mu_l = E(\theta_l)$. Therefore, we have the required result when

$$\prod_{l=1}^{\infty} \left\{ 1 + \mu_l^{1/2} / \delta_l^{3/2} \right\} < \infty.$$

This holds when $\sum_l \mu_l^{1/2} / \delta_l^{3/2} < \infty$. Since we can take $\delta_l \propto l^{-l-r}$ for any $r > 0$ we can take $\mu_l \propto l^{-5-r}$ for any $r > 0$.

5. Discussion

We have provided simple conditions under which a class of regression problems is consistent. The extension to cover more general cases, when f is taken to be a random density function, deserves consideration. Such a model

could provide the likelihood function

$$\prod_{i=1}^n f(y_i - \theta(x_i))$$

with f a random symmetric (about 0) density function. The prior would then be of the type $\Pi(df, d\theta) = \Pi_f(df) \Pi_\theta(d\theta)$ if f and θ are assumed independent a priori. In fact, much of the theory should go through pretty much in the same way as presented in this paper.

It would also be appropriate to study rates of convergence. This would be prior specific and, for example, for the normal model, would involve a refinement of the material appearing in Section 3. Briefly, for the normal model and with Θ compact, we consider $\varepsilon_n \downarrow 0$; so that

$$\varepsilon_n^* = 1 - \exp\{-\varepsilon_n^2/(8\sigma^2)\}$$

and choose $\delta_n = \frac{1}{2}\varepsilon_n$ so that

$$\delta_n^* = 1 - \exp\{-\varepsilon_n^2/(32\sigma^2)\}.$$

Consequently, let

$$\gamma_n = \exp\{-\varepsilon_n^2/(32\sigma^2)\} - \exp\{-\varepsilon_n^2/(8\sigma^2)\}.$$

A straightforward replication of Theorem 1 provides

$$\Pi_n(A_{\varepsilon_n}) \leq I_n^{-1/2} \exp(-nb_n) \text{ a.s. for all large } n$$

for sequences b_n satisfying

$$\sum_n \exp(nb_n) N_{\delta_n} (1 - \gamma_n)^n < \infty.$$

An almost sure (for large n) lower bound of the form $I_n > \exp(-nt_n)$ for some sequence $t_n \downarrow 0$ is typically available; see Shen and Wasserman (2001). Here N_{δ_n} is the (finite) number of Hellinger balls of size δ_n which cover Θ .

Appendix A1.

We will be interested in

$$A_\varepsilon = \bigcup_{h(f_k, f_0) > \varepsilon} A_k.$$

It is clear that A_ε will be a Hellinger neighbourhood of f_0 which shrinks to f_0 as $\varepsilon \downarrow 0$. Then

$$\Pi_n(A_\varepsilon) = \sum_{h(f_k, f_0) > \varepsilon} \Pi_n(A_k).$$

Writing $f_{nA_k} \equiv f_{nk}$, we have

$$h(f_{nA_k}, f_k) \leq \varepsilon/2$$

and by the triangle inequality we have

$$h(f_{nk}, f_0) \geq h(f_k, f_0) - h(f_{nk}, f_k) \geq \varepsilon - \varepsilon/2 = \varepsilon/2.$$

Then, for any $0 < \alpha < 1$, we have

$$\sum_{h(f_k, f_0) > \varepsilon} \Pi_n(A_k) \leq \sum_{h(f_k, f_0) > \varepsilon} \Pi_n(A_k)^\alpha = \sum_{h(f_k, f_0) > \varepsilon} L_{nk}^\alpha / I_n^\alpha,$$

where $I_{nk} \equiv I_{nA_k}$. The Kullback–Leibler property guarantees that $I_n^\alpha > \exp(-nc)$ almost surely for all large n for any $c > 0$. Hence,

$$\Pi_n(A_\varepsilon) \leq \exp(nc) \sum_{h(f_k, f_0) > \varepsilon} L_{nk}^\alpha$$

almost surely for all large n and for any $c > 0$. Now, for $\alpha \geq \frac{1}{2}$, using (1) we have

$$\begin{aligned}
E(L_{n+1k}^\alpha | \mathcal{F}_n) &= L_{nk}^\alpha \int f_{nk}^\alpha f_0^{1-\alpha} \\
&= L_{nk}^\alpha \int (f_0/f_{nk})^{1-\alpha} f_{nk} \\
&\leq L_{nk}^\alpha \left\{ \int (f_0/f_{nk})^{1/2} f_{nk} \right\}^{2(1-\alpha)} \\
&\leq L_{nk}^\alpha \{1 - \varepsilon/2\}^{2(1-\alpha)}.
\end{aligned}$$

Here $\mathcal{F}_n = \sigma(Y_1, \dots, Y_n)$. Clearly then,

$$E(L_{nk}^\alpha) \leq (1 - \varepsilon/2)^{n2(1-\alpha)} \Pi(A_k)^\alpha$$

and so

$$\begin{aligned}
&\sum_n P \left\{ \sum_{d(f_k, f_0) > \varepsilon} L_{nk}^\alpha > \exp(-nb) \right\} \\
&< \sum_n \sum_k \exp(nb) (1 - \varepsilon/2)^{n2(1-\alpha)} \Pi(A_k)^\alpha.
\end{aligned}$$

Selecting $b < -2(1 - \alpha) \log(1 - \varepsilon/2)$; if

$$\sum_k \Pi(A_k)^\alpha < \infty$$

then Borel-Cantelli applies and we have

$$\Pi_n(A_\varepsilon) < \exp(nc - nb)$$

almost surely for all large n with $c < b$. This completes the proof.

Appendix A2.

We prove this theorem by first assuming Θ is compact and then deal with the extension from compactness to separability. With the assumption of compactness, for each $\delta > 0$, there exists $N_\delta < \infty$ and $\{\theta_1 \dots \theta_{N_\delta}\}$ such that

$$\Theta = \bigcup_{k=1}^{N_\delta} A_k$$

and $A_k = \{\theta : d(\theta_k, \theta) < \delta\}$. We consider only those A_k for which $d(\theta_0, \theta_k) > \varepsilon$ which, by (A2), means there exists an ε^* such that $h(f(\cdot|\theta_0(x)), f(\cdot|\theta_k(x))) > \varepsilon^*$ for all x and all k . If $\theta \in A_k$ then by (A1) there exists a δ^* such that

$$h(f(\cdot|\theta(x)), f(\cdot|\theta_k(x))) < \delta^*$$

for all x and all k . Hence, using the convexity of h , one has

$$h(f_{nA_k}, f_{nk}) \leq \int_{A_k} h(f(\cdot|\theta(X_{n+1})), f(\cdot|\theta_k(X_{n+1}))) \Pi_{nA_k}(d\theta) \leq \delta^*,$$

where $f_{nk}(y) = f(y|\theta_k(X_{n+1}))$. Then

$$\begin{aligned} h(f_{nA_k}, f_{n\theta_0}) &\geq h(f_{nk}, f_{n\theta_0}) - h(f_{nk}, f_{nA_k}) \\ &\geq \varepsilon^* - \delta^*. \end{aligned}$$

Let us write $\gamma = \varepsilon^* - \delta^*$ and we will choose δ to ensure that $\gamma > 0$. In this case, using (2), we have

$$E\left\{J_n^{1/2}(A_k)\right\} < \sqrt{\Pi(A_k)}(1-\gamma)^n$$

and so

$$E\left\{\sum_{d(\theta_0, \theta_k) > \varepsilon} J_n^{1/2}(A_k)\right\} < (1-\gamma)^n N_\delta.$$

Therefore, an elementary application of Borel-Cantelli, yields

$$\sum_{d(\theta_0, \theta_k) > \varepsilon} J_n^{1/2}(A_k) < \exp(-nb)$$

a.s. for all large n , for $b < -\log(1-\gamma)$. Now, if

$$A_\varepsilon = \bigcup_{d(\theta_0, \theta_k) > \varepsilon} A_k$$

then

$$\begin{aligned}\Pi_n(A_\varepsilon) &\leq \sum_{d(\theta_0, \theta_k) > \varepsilon} \sqrt{\Pi_n(A_k)} \\ &= I_n^{-1/2} \sum_{d(\theta_0, \theta_k) > \varepsilon} J_n^{1/2}(A_k) \\ &\leq I_n^{-1/2} \exp(-nb) \text{ a.s. for all large } n.\end{aligned}$$

The Kullback-Leibler property result completes the proof for Θ compact.

For a separable space Θ , the proof is the same with the obvious modification;

$$E \left\{ \sum_{d(\theta_0, \theta_k) > \varepsilon} J_n^{1/2}(A_k) \right\} < (1 - \gamma)^n \sum_{d(\theta_0, \theta_k) > \varepsilon} \sqrt{\Pi(A_k)}.$$

Appendix A3.

Let us write $\theta_2 = \theta_1 + \varepsilon$ and put $\theta = \theta_1$, so that we are interested in

$$\begin{aligned}1 - \sqrt{(1 - \theta)^2 - \varepsilon(1 - \theta)} - \sqrt{\theta^2 + \varepsilon\theta} \\ = 1 - (1 - \theta)\sqrt{1 - \varepsilon/(1 - \theta)} - \theta\sqrt{1 + \varepsilon/\theta}.\end{aligned}$$

Without loss of generality we assume that $\varepsilon > 0$. In which case

$\sqrt{1 - \varepsilon/(1 - \theta)} \geq 1 - \varepsilon/(1 - \theta)$ and $\sqrt{1 + \varepsilon/\theta} \geq 1$ so that

$$1 - (1 - \theta)\sqrt{1 - \varepsilon/(1 - \theta)} - \theta\sqrt{1 + \varepsilon/\theta} \leq \varepsilon.$$

Also, $\sqrt{1 - \varepsilon/\theta} < 1$ and $\sqrt{1 + \varepsilon/\theta} < 1 + \frac{1}{2}\varepsilon/\theta$ so that

$$1 - (1 - \theta)\sqrt{1 - \varepsilon/(1 - \theta)} - \theta\sqrt{1 + \varepsilon/\theta} \geq \frac{1}{2}\varepsilon.$$

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